

PROBLEMS Week: 30 April

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

OQ9.4, OQ9.5, OQ9.9, OQ9.10, OQ9.16, CQ9.1, CQ9.3, CQ9.5, CQ9.12, CQ9.13

P9.5 $\vec{F}_{\text{bat}} = (3.26\hat{i} - 3.99\hat{j}) \text{ kN}$

P9.11 (a) -6.00 m/s (b) 8.40 J (c-g) Explanations

P9.17 (a) $9.60 \times 10^{-2} \text{ s}$ (b) $3.65 \times 10^5 \text{ N}$ (c) 26.5 g

P9.27 (a) Fraction of kinetic energy transferred is 0.284 (b) $K_n = 1.15 \times 10^{-13} \text{ J}$ and $K_C = 4.54 \times 10^{-14} \text{ J}$

P9.30 $v = \frac{4M}{m} \sqrt{g\ell}$

P9.33 0.556 m

P9.39 $v_Y = v_i \sin \theta$ and $v_O = v_i \cos \theta$

P9.44 $v_{\text{blue}} = 5.89 \text{ m/s}$ and $v_{\text{green}} = 7.07 \text{ m/s}$

P9.48 $\vec{r}_{\text{CM}} = (11.7\hat{i} + 13.3\hat{j}) \text{ cm}$

P9.50 $\vec{r}_{\text{CM}} = (-0.0067\hat{j}) \text{ nm}$ if the oxygen atom is at the origin.

P9.52 (a) Sketch (b) $\vec{r}_{\text{CM}} = (-2.00\hat{i} - 1.00\hat{j}) \text{ m}$ (c) $\vec{v}_{\text{CM}} = (3.00\hat{i} - 1.00\hat{j}) \text{ m/s}$

(d) $\vec{P} = (15.0\hat{i} - 5.00\hat{j}) \text{ kg}\cdot\text{m/s}$

P9.67 $\vec{F}_{\text{avg}} = -260\hat{i} \text{ N}$

P9.80 (a) $v_{\text{wedge}} = -0.667 \text{ m/s}$ (b) $h = 0.952 \text{ m}$

P9.81 $v_i = 143 \text{ m/s}$